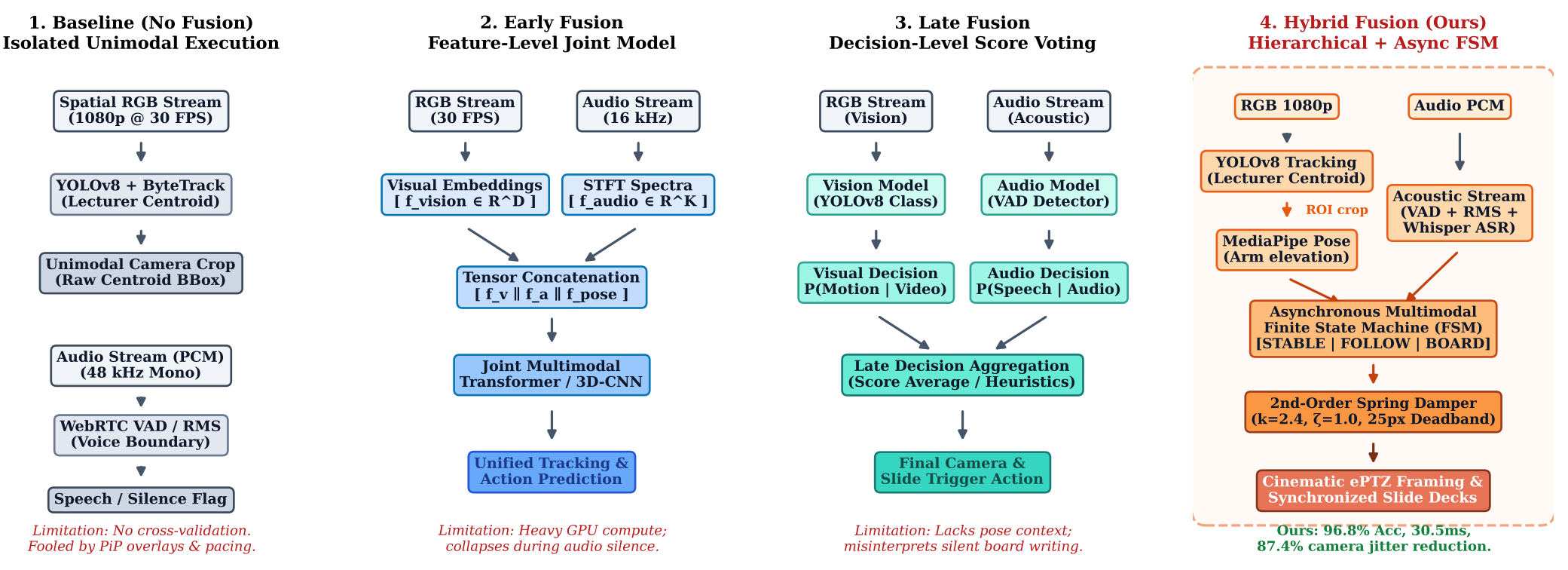
1. Baseline (No Fusion)
Isolated Unimodal Execution
Spatial RGB Stream (1080p @ 30 FPS)
YOLOv8 + ByteTrack (Lecturer Centroid)
Unimodal Camera Crop (Raw Centroid BBox)
Audio Stream (PCM) (48 kHz Mono)
WebRTC VAD / RMS (Voice Boundary)
Speech / Silence Flag
Limitation: No cross-validation.
Fooled by PiP overlays & pacing.
2. Early Fusion
Feature-Level Joint Model
RGB Stream (30 FPS)
Audio Stream (16 kHz)
Visual Embeddings [f_vision ∈ R^D]
STFT Spectra [f_audio ∈ R^K]
Tensor Concatenation [f_v ‖ f_a ‖ f_pose]
Joint Multimodal Transformer / 3D-CNN
Unified Tracking & Action Prediction
Limitation: Heavy GPU compute;
collapses during audio silence.
3. Late Fusion
Decision-Level Score Voting
RGB Stream (Vision)
Audio Stream (Acoustic)
Vision Model (YOLOv8 Class)
Audio Model (VAD Detector)
Visual Decision P(Motion | Video)
Audio Decision P(Speech | Audio)
Late Decision Aggregation (Score Average / Heuristics)
Final Camera & Slide Trigger Action
Limitation: Lacks pose context;
misinterprets silent board writing.
4. Hybrid Fusion (Ours)
Hierarchical + Async FSM
RGB 1080p
Audio PCM
YOLOv8 Tracking (Lecturer Centroid)
ROI crop
MediaPipe Pose (Arm elevation)
Acoustic Stream (VAD + RMS + Whisper ASR)
Asynchronous Multimodal Finite State Machine (FSM) [STABLE | FOLLOW | BOARD]
2nd-Order Spring Damper (k=2.4, ζ=1.0, 25px Deadband)
Cinematic ePTZ Framing & Synchronized Slide Decks
Ours: 96.8% Acc, 30.5ms,
87.4% camera jitter reduction.